

Setting the environment

```
C:\ILMENAU\cc\apl>python -m venv venv
C:\ILMENAU\cc\apl>venv\Scripts\activate

(venv) C:\ILMENAU\cc\apl>pip install -r requirements.txt
Requirement already satisfied: fastapi in .\venv\Lib\site-packages (from -r requirements.txt (line 1)) (0.136.3)
Requirement already satisfied: pillow in .\venv\Lib\site-packages (from -r requirements.txt (line 3)) (12.2.0)
Requirement already satisfied: python-multipart in .\venv\Lib\site-packages (from -r requirements.txt (line 4)) (0.0.32)
Requirement already satisfied: torch in .\venv\Lib\site-packages (from -r requirements.txt (line 5)) (2.12.0)
Requirement already satisfied: torchvision in .\venv\Lib\site-packages (from -r requirements.txt (line 6)) (0.27.0)
Requirement already satisfied: uvicorn[standard] in .\venv\Lib\site-packages (from -r requirements.txt (line 2)) (0.49.0)
Requirement already satisfied: starlette>=0.46.0 in .\venv\Lib\site-packages (from fastapi->r requirements.txt (line 1)) (1.2.1)
Requirement already satisfied: pydantic>=2.9.0 in .\venv\Lib\site-packages (from fastapi->r requirements.txt (line 1)) (2.13.4)
Requirement already satisfied: typing-extensions>=4.8.0 in .\venv\Lib\site-packages (from fastapi->r requirements.txt (line 1)) (4.15.0)
Requirement already satisfied: typing-inspection>=0.4.2 in .\venv\Lib\site-packages (from fastapi->r requirements.txt (line 1)) (0.4.2)
Requirement already satisfied: annotated-doc>=0.0.2 in .\venv\Lib\site-packages (from fastapi->r requirements.txt (line 1)) (0.0.4)
Requirement already satisfied: click>=7.0 in .\venv\Lib\site-packages (from uvicorn[standard]->r requirements.txt (line 2)) (8.4.1)
Requirement already satisfied: h11>=0.8 in .\venv\Lib\site-packages (from uvicorn[standard]->r requirements.txt (line 2)) (0.16.0)
Requirement already satisfied: colorama>=0.4 in .\venv\Lib\site-packages (from uvicorn[standard]->r requirements.txt (line 2)) (0.4.6)
Requirement already satisfied: httptools>=0.8.0 in .\venv\Lib\site-packages (from uvicorn[standard]->r requirements.txt (line 2)) (0.8.0)
Requirement already satisfied: python-dotenv>=0.13 in .\venv\Lib\site-packages (from uvicorn[standard]->r requirements.txt (line 2)) (1.2.2)
Requirement already satisfied: pyyaml>=5.1 in .\venv\Lib\site-packages (from uvicorn[standard]->r requirements.txt (line 2)) (6.0.3)
Requirement already satisfied: watchfiles>=0.20 in .\venv\Lib\site-packages (from uvicorn[standard]->r requirements.txt (line 2)) (1.2.0)
Requirement already satisfied: websockets>=10.4 in .\venv\Lib\site-packages (from uvicorn[standard]->r requirements.txt (line 2)) (16.0)
Requirement already satisfied: filelock in .\venv\Lib\site-packages (from torch->r requirements.txt (line 5)) (3.29.3)
Requirement already satisfied: setuptools<82 in .\venv\Lib\site-packages (from torch->r requirements.txt (line 5)) (65.5.0)
Requirement already satisfied: sympy>=1.13.3 in .\venv\Lib\site-packages (from torch->r requirements.txt (line 5)) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in .\venv\Lib\site-packages (from torch->r requirements.txt (line 5)) (3.4.2)
Requirement already satisfied: Jinja2 in .\venv\Lib\site-packages (from torch->r requirements.txt (line 5)) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in .\venv\Lib\site-packages (from torchvision->r requirements.txt (line 6)) (2026.4.0)
Requirement already satisfied: numpy in .\venv\Lib\site-packages (from torchvision->r requirements.txt (line 6)) (2.2.6)
Requirement already satisfied: annotated-types>=0.6.0 in .\venv\Lib\site-packages (from pydantic>=2.9.0->fastapi->r requirements.txt (line 1)) (0.7.0)
Requirement already satisfied: pydantic-core>=2.46.4 in .\venv\Lib\site-packages (from pydantic>=2.9.0->fastapi->r requirements.txt (line 1)) (2.46.4)
Requirement already satisfied: anyio<5,>=3.6.2 in .\venv\Lib\site-packages (from starlette>=0.46.0->fastapi->r requirements.txt (line 1)) (4.13.0)
Requirement already satisfied: idna>=2.8 in .\venv\Lib\site-packages (from anyio<5,>=3.6.2->starlette>=0.46.0->fastapi->r requirements.txt (line 1)) (3.18)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in .\venv\Lib\site-packages (from sympy>=1.13.3->torch->r requirements.txt (line 5)) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in .\venv\Lib\site-packages (from Jinja2->torch->r requirements.txt (line 5)) (3.0.3)

[notice] A new release of pip is available: 26.1.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
```

Running the API Locally

```
(venv) C:\ILMENAU\cc\apl>python -m uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\\ILMENAU\\cc\\apl']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [20664] using WatchFiles
INFO: Started server process [26400]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [26400]
INFO: Stopping reloader process [20664]
```

FastAPI 0.10.0 OAS 3.1

/openapi.json

default	^
POST /predict Predict	v
GET /health Health	v
Schemas	^
Body_predict_predict_post > Expand all object	
HTTPValidationError > Expand all object	
ValidationError > Expand all object	

Building the Docker Image

```
(venv) C:\ILMENAU\cc\apl>kubectl get nodes
NAME          STATUS    ROLES    AGE   VERSION
minikube      Ready     control-plane  12h   v1.35.1

(venv) C:\ILMENAU\cc\apl>docker build -t imagenet-api .
[+] Building 576.3s (12/12) FINISHED                                docker:desktop-linux
=> [internal] load build definition from dockerfile                 0.0s
=> => transferring dockerfile: 396B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.10-slim 2.5s
=> [auth] library/python:pull token for registry-1.docker.io       0.0s
=> [internal] load .dockerignore                                    0.0s
=> => transferring context: 66B                                     0.0s
=> [1/6] FROM docker.io/library/python:3.10-slim@sha256:1cd276d6838b4e249d57e904e36969465a3523772d14f0222743a95607f7fc30 156.2s
=> => resolve docker.io/library/python:3.10-slim@sha256:1cd276d6838b4e249d57e904e36969465a3523772d14f0222743a95607f7fc30 0.1s
=> => sha256:761b88a3de45a6f5174e0c4d1df0d0e914f455f182840c27d50ddc4d45fb88a4 249B / 249B 0.2s
=> => sha256:3c58578f39599387ef9d2d5ba89b53cfc4e1015508e73dc63d008b672d95bc9 13.82MB / 13.82MB 108.6s
=> => sha256:f41c8231c824bc5b8565006225367cfff6d5d78dd84a6292a459566a78f4ad0cc 1.29MB / 1.29MB 5.6s
=> => sha256:72c03230f1363a3fb61d2f98504cf168bad3fe22f511ad2005dc021515d7ce97 29.79MB / 29.79MB 153.4s
=> => extracting sha256:72c03230f1363a3fb61d2f98504cf168bad3fe22f511ad2005dc021515d7ce97 1.5s
=> => extracting sha256:f41c8231c824bc5b8565006225367cfff6d5d78dd84a6292a459566a78f4ad0cc 0.2s
=> => extracting sha256:3c58578f39599387ef9d2d5ba89b53cfc4e1015508e73dc63d008b672d95bc9 0.9s
=> => extracting sha256:761b88a3de45a6f5174e0c4d1df0d0e914f455f182840c27d50ddc4d45fb88a4 0.0s
=> [internal] load build context                                    0.0s
=> => transferring context: 830B                                    0.0s
=> [2/6] WORKDIR /app                                             0.2s
=> [3/6] RUN apt-get update && apt-get install -y libjpeg-dev zlibg-dev && rm -rf /var/lib/apt/lists/* 13.3s
=> [4/6] COPY requirements.txt .                                  0.1s
=> [5/6] RUN pip install --default-timeout=1000 --no-cache-dir -r requirements.txt 200.9s
```

Verify the Docker Image

```
(venv) C:\ILMENAU\cc\apl>docker images
```

IMAGE	ID	DISK USAGE	Info →	U	In Use
			CONTENT SIZE	SIZE	EXTRA
docker/desktop-cloud-provider-kind:v0.5.0	4ad59ce20658	595MB		162MB	U
docker/desktop-containerd-registry-mirror:v0.0.3	179a9ea3ae47	50.3MB		16.1MB	U
dog-breed-api:latest	f73d894e105d	13.3GB		4.19GB	U
envoyproxy/envoy:v1.36.4	08acc53db6f4	248MB		62.1MB	
gcr.io/k8s-minikube/kicbase:v0.0.50	b97074569ae9	1.94GB		545MB	
gcr.io/k8s-minikube/kicbase@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93	eb4fec00e8ad	1.94GB		545MB	U
imagenet-api:latest	b308738a885c	8.1GB		2.87GB	
kicbase/stable:v0.0.50	b97074569ae9	1.94GB		545MB	
kicbase/stable@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93	eb4fec00e8ad	1.94GB		545MB	U
kindest/node:v1.34.3	08497ee19eac	1.35GB		395MB	U
ml-inference-server:latest	56bf7c65424e	8.1GB		2.87GB	U
sagarkumar9/dispatcher:latest	b7316664812e	219MB		53MB	
sagarkumar9/dog-breed-api:latest	f73d894e105d	13.3GB		4.19GB	U
sagarkumar9/ml-inference-server:latest	56bf7c65424e	8.1GB		2.87GB	U

Run Container

```
(venv) C:\ILMENAU\cc\apl>docker run -p 8080:8080 imagenet-api
100.0%
INFO:      Started server process [1]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
INFO:      Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
Downloading: "https://download.pytorch.org/models/resnet18-f37072fd.pth" to /root/.cache/torch/hub/checkpoints/resnet18-f37072fd.pth
INFO:      172.17.0.1:44968 - "GET /docs HTTP/1.1" 200 OK
INFO:      172.17.0.1:44968 - "GET /openapi.json HTTP/1.1" 200 OK
INFO:      Shutting down
INFO:      Waiting for application shutdown.
INFO:      Application shutdown complete.
INFO:      Finished server process [1]
```

Load image into Minikube and Deployment

```
(venv) C:\ILMENAU\cc\apl>minikube image load imagenet-api:latest

(venv) C:\ILMENAU\cc\apl>minikube image ls | findstr imagenet
docker.io/library/imagenet-api:latest

(venv) C:\ILMENAU\cc\apl>kubectl apply -f deployment.yaml
deployment.apps/ml-inference created

(venv) C:\ILMENAU\cc\apl>kubectl apply -f service.yaml
service/ml-inference-service created

(venv) C:\ILMENAU\cc\apl>kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
ml-inference-6457777ff9-wkcxh  1/1     Running   0           56s

(venv) C:\ILMENAU\cc\apl>minikube service ml-inference-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	ml-inference-service	8080	http://192.168.49.2:30080

* Starting tunnel for service ml-inference-service.

NAMESPACE	NAME	TARGET PORT	URL
default	ml-inference-service		http://127.0.0.1:64742

* Opening service default/ml-inference-service in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
* Stopping tunnel for service ml-inference-service.

Install Load Tester Locust and Run

```
(venv) C:\ILMENAU\cc\apl>locust --version
locust 2.44.1 from C:\ILMENAU\cc\apl\venv\Lib\site-packages\locust (Python 3.14.5)
Exception ignored on calling weakref callback <function _removeHandlerRef at 0x0000017515857950>:
Traceback (most recent call last):
  File "C:\Users\B SAGAR KUMAR\AppData\Local\Python\pythoncore-3.14-64\Lib\logging\__init__.py", line 893, in _removeHandlerRef
  File "C:\Users\B SAGAR KUMAR\AppData\Local\Python\pythoncore-3.14-64\Lib\threading.py", line 207, in acquire
  File "C:\ILMENAU\cc\apl\venv\Lib\site-packages\gevent\thread.py", line 81, in get_ident
RuntimeError: greenlet is being finalized

(venv) C:\ILMENAU\cc\apl>locust -f locustfile.py --host=http://localhost:8080
[2026-06-11 13:21:12,808] LAPTOP-ROLIEI6G/INFO/locust.main: Starting Locust 2.44.1
[2026-06-11 13:21:12,809] LAPTOP-ROLIEI6G/INFO/locust.main: Starting web interface at http://localhost:8089, press enter to open your default browser.
[2026-06-11 13:21:57,400] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 20 users at a rate of 5.00 per second
[2026-06-11 13:22:00,438] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 20} (20 total users)
Traceback (most recent call last):
  File "C:\ILMENAU\cc\apl\venv\Lib\site-packages\gevent\_ffi\loop.py", line 279, in python_check_callback
    def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument

KeyboardInterrupt
2026-06-11 13:31:31Z
[2026-06-11 13:31:31,024] LAPTOP-ROLIEI6G/INFO/locust.main: Shutting down (exit code 1)
Type      Name      # reqs      # fails      Avg      Min      Max      Med      req/s      failures/s
-----
POST      /predict  78          25(32.05%)   103841    10767    393257    39000    0.15       0.05
-----
Aggregated 78          25(32.05%)   103841    10767    393257    39000    0.15       0.05
-----

Response time percentiles (approximated)
Type      Name      50%      66%      75%      80%      90%      95%      98%      99%      99.9%      99.99%      100% # reqs
-----
POST      /predict  39000    47000    159000    251000    337000    356000    375000    393000    393000    393000    393000 78
-----
Aggregated 39000    47000    159000    251000    337000    356000    375000    393000    393000    393000    393000 78
-----

Error report
# occurrences      Error
-----
25                 POST /predict: RemoteDisconnected('Remote end closed connection without response')
-----
```

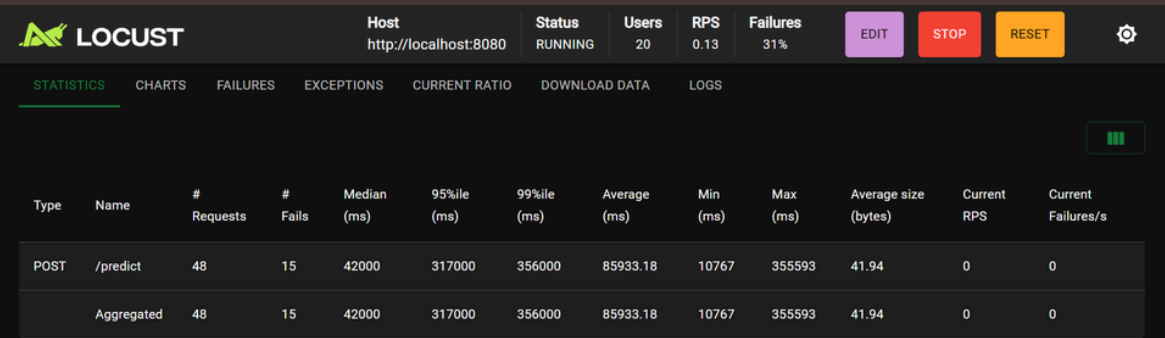
Start Port Forward

```
(venv) C:\ILMENAU\cc\apl>kubectl port-forward service/ml-inference-service 8080:8080
Forwarding from 127.0.0.1:8080 -> 8080
Forwarding from [::1]:8080 -> 8080
Handling connection for 8080
Handling connection for 8080
```

Check

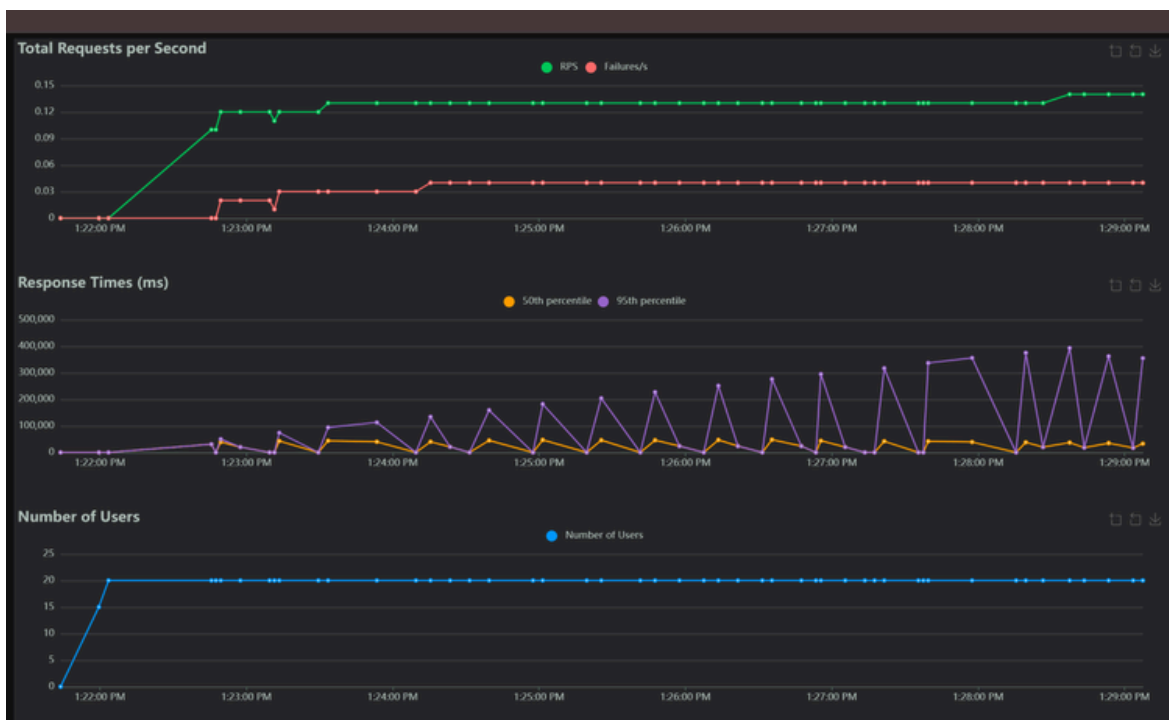
```
C:\ILMENAU\cc\apl>kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
ml-inference-86f7d8bcd4-2pvmz      1/1     Running   0           25m
```

Statistics



Failures

LOCUST				Host	Status	Users	RPS	Failures	EDIT	STOP	RESET	
				http://localhost:8080	RUNNING	20	0.15	32%				
STATISTICS	CHARTS	FAILURES	EXCEPTIONS	CURRENT RATIO	DOWNLOAD DATA	LOGS						
# Failures	Method	Name	Message					First Seen		Last Seen		
23	POST	/predict	RemoteDisconnected('Remote end closed connection without response')					6/11/2026, 1:22:48 PM		6/11/2026, 1:30:06 PM		



Build the Dispatcher image and Verify

```
(venv) C:\ILMENAU\cc\apl>docker build -t dispatcher:latest ./dispatcher
[+] Building 83.6s (11/11) FINISHED
=> [internal] load build definition from dockerfile
=> => transferring dockerfile: 238B
=> [internal] load metadata for docker.io/library/python:3.11-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 4.54kB
[1/5] FROM docker.io/library/python:3.11-slim@sha256:f9fa7f851e38bf19c9de3afbc4b86ae7176ea7aaf94535c31df5458d5849457
=> resolve docker.io/library/python:3.11-slim@sha256:f9fa7f851e38bf19c9de3afbc4b86ae7176ea7aaf94535c31df5458d5849457
=> sha256:8bd2cf7d3f6acc894d6ab00affeb6abf779933fa982aec49b9a4e322b7fbb 14.36MB / 14.36MB
=> sha256:3258ea7ceadcle9a40a2d832082ee8fe68631ef9c9bf7bd5447fbcff9f527439 250B / 250B
=> sha256:8c471a93fc525f2012a1d483a12540ba9df9189f0bfe12238f74273aa72590a7 1.29MB / 1.29MB
=> extracting sha256:8c471a93fc525f2012a1d483a12540ba9df9189f0bfe12238f74273aa72590a7
=> extracting sha256:8bd2cf7d3f6acc894d6ab00affeb6abf779933fa982aec49b9a4e322b7fbb
=> extracting sha256:3258ea7ceadcle9a40a2d832082ee8fe68631ef9c9bf7bd5447fbcff9f527439
[2/5] WORKDIR /app
[3/5] COPY requirements.txt .
[4/5] RUN pip install --no-cache-dir -r requirements.txt
[5/5] COPY . .
=> exporting to image
=> exporting layers
=> exporting manifest sha256:70860beee49822ae639293997881327dd4c339615d8fc983288def9ce65ab975
=> exporting config sha256:031e55dbe281cabea7c2baab1f18a6aa343cfaec8bd3f3f5242a444dd0b93ed6
=> exporting attestation manifest sha256:39569f86729e519a02293c815f6527cc337d5e6e7183666c21c5a0799e57cab9
=> exporting manifest list sha256:a5d2567dbceda93f2cd657ce26ad673f3d96b0f3a394a1b56d810c2d954a1889
=> naming to docker.io/library/dispatcher:latest
=> unpacking to docker.io/library/dispatcher:latest
1.1s

(venv) C:\ILMENAU\cc\apl>docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	Info →	U In Use
dispatcher:latest	a5d2567dbced	229MB	55.3MB		
docker/desktop-cloud-provider-kind:v0.5.0	4ad59ce20658	595MB	162MB		U
docker/desktop-containernerd-registry-mirror:v0.0.3	179a9ea3ae47	50.3MB	16.1MB		U
dog-breed-api:latest	f73d894e105d	13.3GB	4.19GB		U
envoyproxy/envoy:v1.36.4	08acc53db6f4	248MB	62.1MB		U
gcr.io/k8s-minikube/kicbase:v0.0.50	b97074569ae9	1.94GB	545MB		U
gcr.io/k8s-minikube/kicbase@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93	eb4fec00e8ad	1.94GB	545MB		U
imagenet-api:latest	757ebdd0899f	8.1GB	2.87GB		U
kicbase/stable:v0.0.50	b97074569ae9	1.94GB	545MB		U
kicbase/stable@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93	eb4fec00e8ad	1.94GB	545MB		U
kindest/node:v1.34.3	08497ee19eac	1.35GB	395MB		U
ml-inference-server:latest	56bf7c65424e	8.1GB	2.87GB		U
sagarkumar9/dispatcher:latest	b7316664812e	219MB	53MB		U
sagarkumar9/dog-breed-api:latest	f73d894e105d	13.3GB	4.19GB		U
sagarkumar9/ml-inference-server:latest	56bf7c65424e	8.1GB	2.87GB		U

Load the Dispatcher image into Minikube and Deployment

```
(venv) C:\ILMENAU\cc\apl>minikube image load dispatcher:latest

(venv) C:\ILMENAU\cc\apl>kubectl apply -f dispatcher\dispatcher-deployment.yaml
deployment.apps/dispatcher created

(venv) C:\ILMENAU\cc\apl>kubectl apply -f dispatcher\dispatcher-service.yaml
service/dispatcher-service created

(venv) C:\ILMENAU\cc\apl>kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
dispatcher-59c44dbcb5-ph7rp	1/1	Running	0	21s
ml-inference-86f7d8bcd4-2pvmz	1/1	Running	0	51m

```
Pretty-print ☐

{"message":"Dispatcher is running"}
```

Port-Forward

```
(venv) C:\ILMENAU\cc\apl>kubectl port-forward service/dispatcher-service 9000:8000
Forwarding from 127.0.0.1:9000 -> 8000
Forwarding from [::1]:9000 -> 8000
Handling connection for 9000
Handling connection for 9000
Handling connection for 9000
Handling connection for 9000
```

Install Metrics Server

```
(venv) C:\ILMENAU\cc\apl>kubectl apply -f https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/components.yaml
serviceaccount/metrics-server created
clusterrole.rbac.authorization.k8s.io/system:aggregated-metrics-reader created
clusterrole.rbac.authorization.k8s.io/system:metrics-server created
rolebinding.rbac.authorization.k8s.io/metrics-server-auth-reader created
clusterrolebinding.rbac.authorization.k8s.io/metrics-server:system:auth-delegator created
clusterrolebinding.rbac.authorization.k8s.io/system:metrics-server created
service/metrics-server created
deployment.apps/metrics-server created
apiservice.apiregistration.k8s.io/v1beta1.metrics.k8s.io created

(venv) C:\ILMENAU\cc\apl>kubectl edit deployment metrics-server -n kube-system
deployment.apps/metrics-server edited

(venv) C:\ILMENAU\cc\apl>kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
coredns-7d764666f9-nbqmk           1/1     Running   3 (15m ago)  20h
etcd-minikube                       1/1     Running   3 (15m ago)  20h
kube-apiserver-minikube             1/1     Running   3 (15m ago)  20h
kube-controller-manager-minikube    1/1     Running   3 (15m ago)  20h
kube-proxy-m4c2c                   1/1     Running   3 (15m ago)  20h
kube-scheduler-minikube             1/1     Running   3 (15m ago)  20h
metrics-server-5f54fb74d9-qm895    0/1     Running   0           19s
metrics-server-b4c746d8b-kql5v     0/1     Running   0           3m33s
storage-provisioner                 1/1     Running   6 (14m ago)  20h

(venv) C:\ILMENAU\cc\apl>kubectl top pods
NAME                                CPU(cores)   MEMORY(bytes)
dispatcher-59c44dbcb5-ph7rp         3m           63Mi
ml-inference-86f7d8bcd4-2pvmz       2m           473Mi
```

Install Prometheus and Check if Pods are Running

```
(venv) C:\ILMENAU\cc\apl>kubectl get pods -l "release=prometheus"
NAME                                READY   STATUS    RESTARTS   AGE
prometheus-kube-prometheus-operator-6c65fd7554-hbcb5  0/1     Running   10 (46s ago)  2d
prometheus-kube-state-metrics-6cc7c56db5-lwh75        1/1     Running   11 (46s ago)  3d
prometheus-prometheus-node-exporter-72xlf              1/1     Running   7 (46s ago)  3d
```

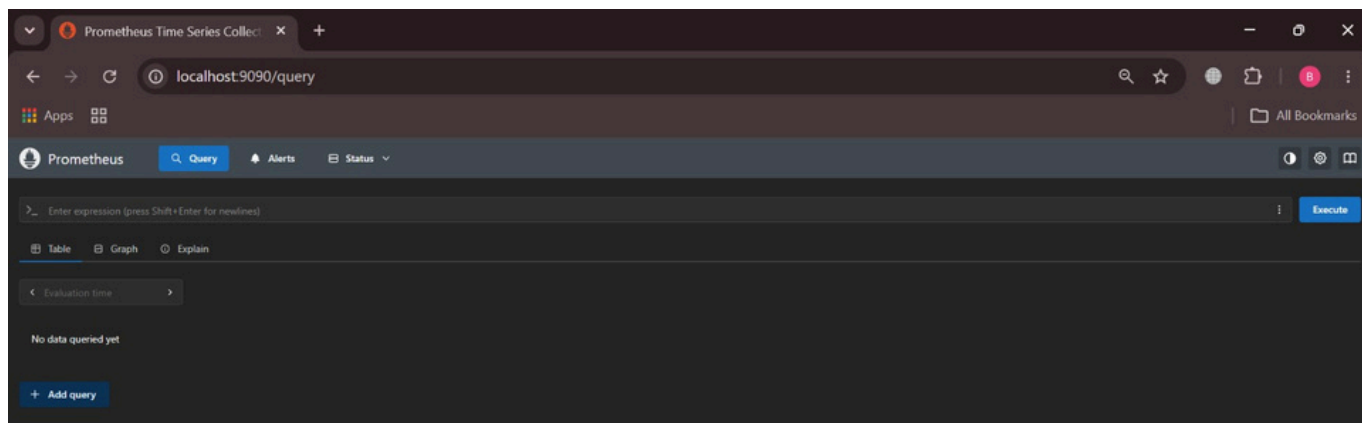
Create ServiceMonitor and Deploy it

```
(venv) C:\ILMENAU\cc\apl>cd dispatcher

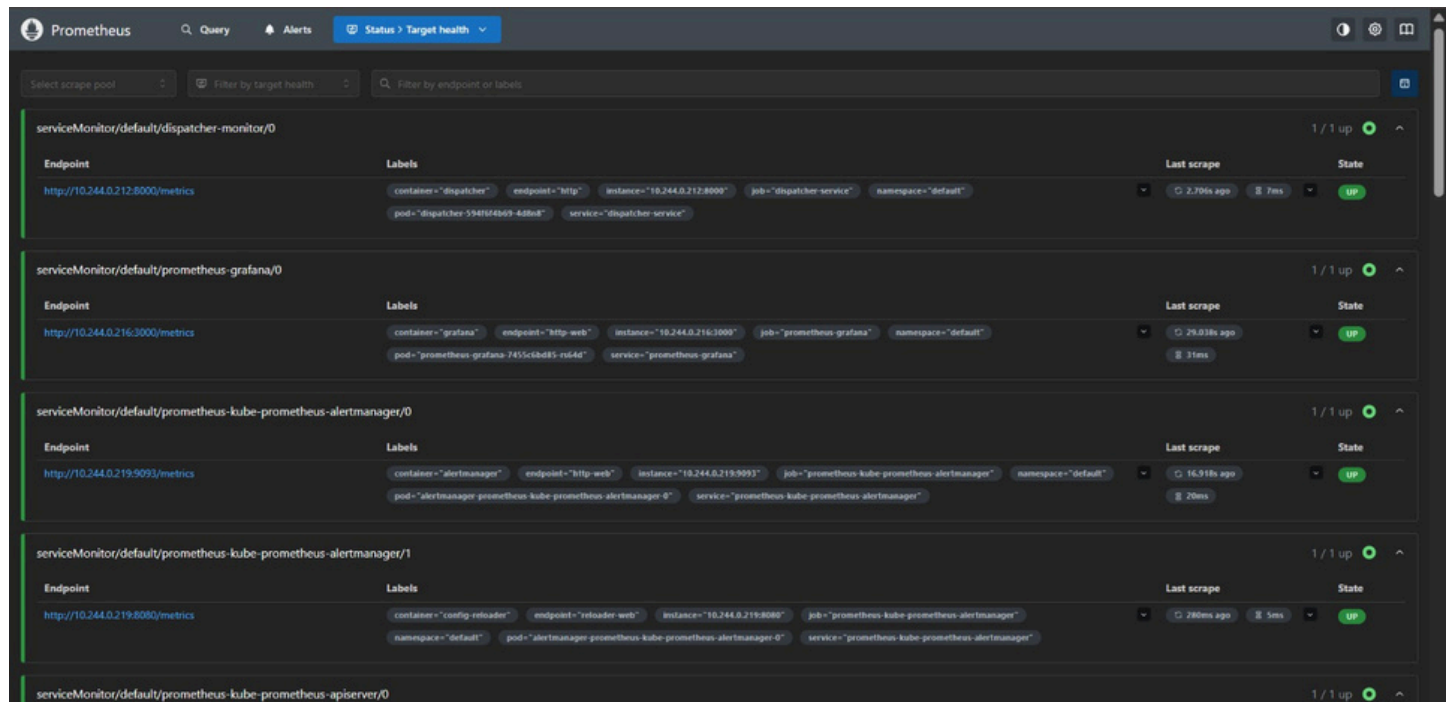
(venv) C:\ILMENAU\cc\apl\dispatcher>kubectl apply -f dispatcher-servicemonitor.yaml
servicemonitor.monitoring.coreos.com/dispatcher-monitor unchanged
```

Port-Forward

```
(venv) C:\ILMENAU\cc\apl\dispatcher>kubectl port-forward svc/prometheus-kube-prometheus-prometheus 9090:9090
Forwarding from 127.0.0.1:9090 -> 9090
Forwarding from [::1]:9090 -> 9090
Handling connection for 9090
Handling connection for 9090
Handling connection for 9090
```



Verify Prometheus is scraping your dispatcher



Build AutoScaler Docker Image

```
(venv) C:\ILMENAU\cc\apl>docker build -t autoscaler:latest .
[+] Building 3.0s (12/12) FINISHED
=> [internal] load build definition from dockerfile
=> => transferring dockerfile: 396B
=> [internal] load metadata for docker.io/library/python:3.10-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 66B
=> [internal] load build context
=> => transferring context: 7.86kB
=> [1/6] FROM docker.io/library/python:3.10-slim@sha256:fa184fce49c170a8b1032a4f752f9fe1a7e463e7f5795a3952ca275e166fa913
=> => resolve docker.io/library/python:3.10-slim@sha256:fa184fce49c170a8b1032a4f752f9fe1a7e463e7f5795a3952ca275e166fa913
=> CACHED [2/6] WORKDIR /app
=> CACHED [3/6] RUN apt-get update && apt-get install -y libjpeg-dev zlib1g-dev && rm -rf /var/lib/apt/lists/*
=> CACHED [4/6] COPY requirements.txt .
=> CACHED [5/6] RUN pip install --default-timeout=1000 --no-cache-dir -r requirements.txt
=> [6/6] COPY .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:52ff1b442b82649b13a63f6c946334d2df1c397e0ee3416f44d39c5d9b5d7589
=> => exporting config sha256:5cf7afafdc9a41bd654719c512ebab7fee491c2fd7f62a341629d349282ee
=> => exporting attestation manifest sha256:8915c922e6409489d55f9342b7e39e6393ff8f1636907dd76b5c5736762c2c50
=> => exporting manifest list sha256:d972ab04303dffc680b1544d5278f45508356b6c95ae27666a3abc2b5eff673
=> => naming to docker.io/library/autoscaler:latest
=> => unpacking to docker.io/library/autoscaler:latest
```

Load the image into Minikube and apply RBAC

```
(venv) C:\ILMENAU\cc\apl>minikube image load autoscaler:latest

(venv) C:\ILMENAU\cc\apl>kubectl apply -f RBAC.yaml
error: the path "RBAC.yaml" does not exist

(venv) C:\ILMENAU\cc\apl>cd autoscaler
The system cannot find the path specified.

(venv) C:\ILMENAU\cc\apl>cd autoscaler

(venv) C:\ILMENAU\cc\apl\autoscaler>kubectl apply -f RBAC.yaml
serviceaccount/autoscaler-sa unchanged
role.rbac.authorization.k8s.io/autoscaler-role unchanged
rolebinding.rbac.authorization.k8s.io/autoscaler-binding unchanged
```

Deploy the AutoScaler

```
(venv) C:\ILMENAU\cc\apl\autoscaler>kubectl apply -f deployment_auto.yaml
deployment.apps/custom-autoscaler unchanged

(venv) C:\ILMENAU\cc\apl\autoscaler>kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
alertmanager-prometheus-kube-prometheus-alertmanager-0	2/2	Running	13 (27m ago)	3d
custom-autoscaler-759475f877-j87nm	1/1	Running	26 (12m ago)	40h
dispatcher-594f6f4b69-4d8n8	1/1	Running	9 (27m ago)	5d17h
ml-inference-5ddfc89bb6-l96mw	1/1	Running	1 (27m ago)	19h
prometheus-grafana-7455c6bd85-rs64d	3/3	Running	18 (27m ago)	3d
prometheus-kube-prometheus-operator-6c65fd7554-hbcbs	1/1	Running	14 (14m ago)	2d
prometheus-kube-state-metrics-6cc7c56db5-lwh75	1/1	Running	14 (13m ago)	3d
prometheus-prometheus-kube-prometheus-prometheus-0	2/2	Running	14 (13m ago)	3d
prometheus-prometheus-node-exporter-72xlf	1/1	Running	8 (13m ago)	3d

Apply 70% HPA

```
C:\Windows\System32\cmd.e x + - - - - -
C:\ILMENAU\cc\apl>kubectl get pods -w
```

NAME	STATUS	RESTARTS	AGE	READY
alertmanager-prometheus-kube-prometheus-alertmanager-0	Running	19 (6m34s ago)	5d1h	2/2
custom-autoscaler-759475f877-rl9bv	Running	5 (4m47s ago)	23h	1/1
dispatcher-594f6f4b69-w7k54	Running	0	3m25s	1/1
ml-inference-6fdb78c657-7cqmw	Running	1 (2m48s ago)	4m31s	1/1
prometheus-grafana-7455c6bd85-rs64d	Running	27 (6m34s ago)	5d1h	3/3
prometheus-kube-prometheus-operator-6c65fd7554-hbcbs	Running	18 (6m34s ago)	4d1h	1/1
prometheus-kube-state-metrics-6cc7c56db5-lwh75	Running	19 (5m18s ago)	5d1h	1/1
prometheus-prometheus-kube-prometheus-prometheus-0	Running	20 (6m34s ago)	5d1h	2/2
prometheus-prometheus-node-exporter-72xlf	Running	11 (6m34s ago)	5d1h	1/1

```
C:\ILMENAU\cc\apl>kubectl apply -f hpa/hpa-70.yaml
horizontalpodautoscaler.autoscaling/ml-inference created
```

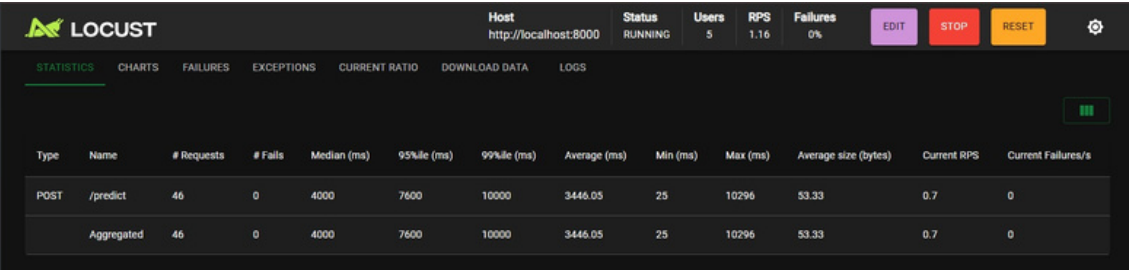
```
C:\Windows\System32\cmd.e x + - - - - -
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\ILMENAU\cc\apl>venv\Scripts\activate

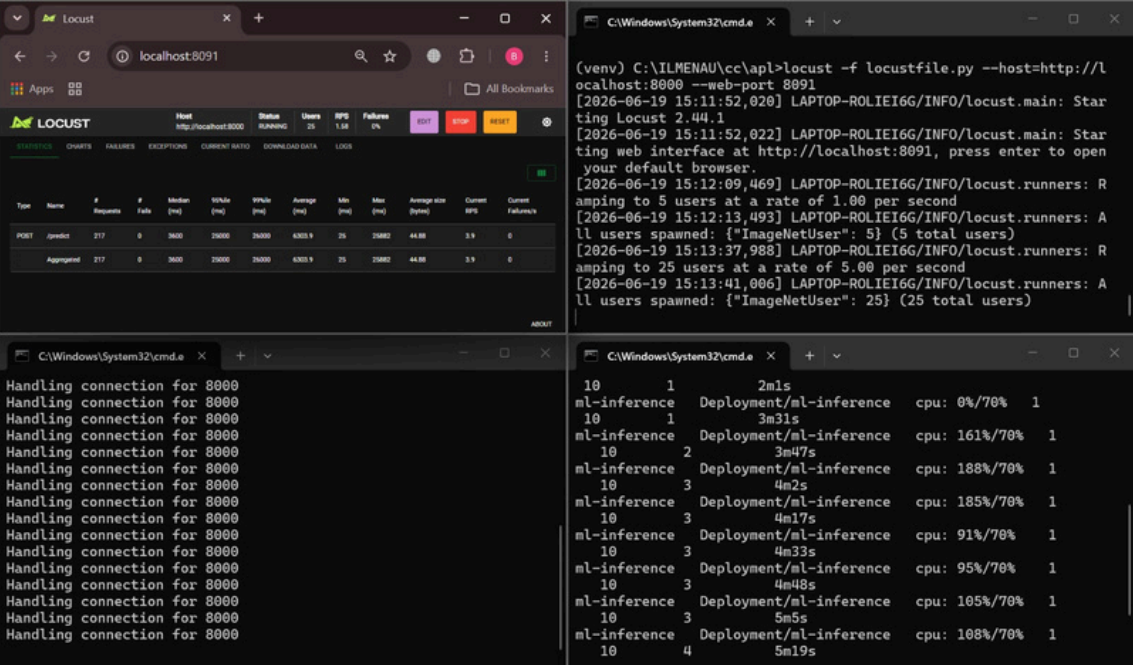
(venv) C:\ILMENAU\cc\apl>kubectl get hpa -w
```

NAME	REFERENCE	AGE	TARGETS	MINPODS
ml-inference	Deployment/ml-inference	cpu: 1%/70%	1	1
ml-inference	Deployment/ml-inference	cpu: 0%/70%	1	1
ml-inference	Deployment/ml-inference	cpu: 1%/70%	1	1

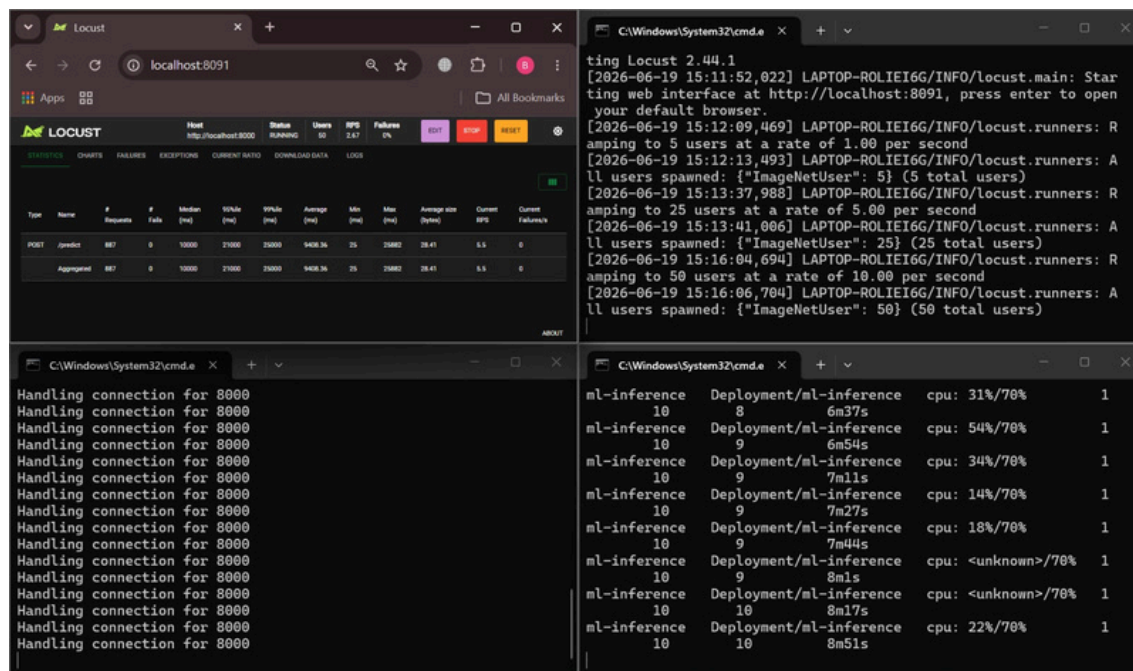
When Load is Less (Users-5, Ram up-1)



When we increase Load (Users-25, Ram up-5)

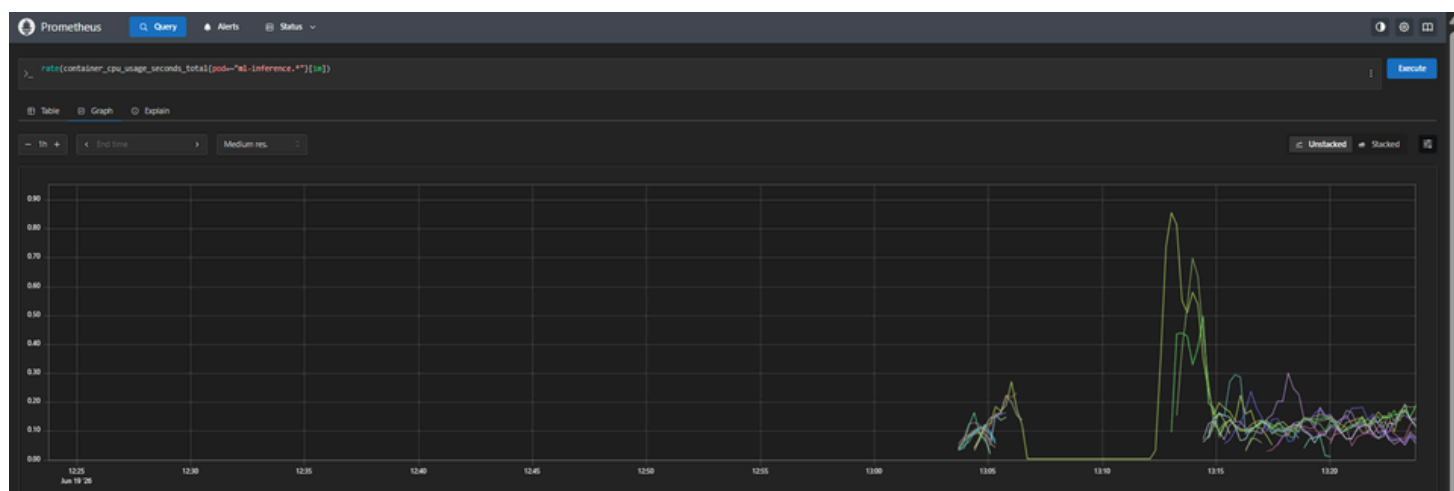


When we increase Load (Users-50, Ram up-10)

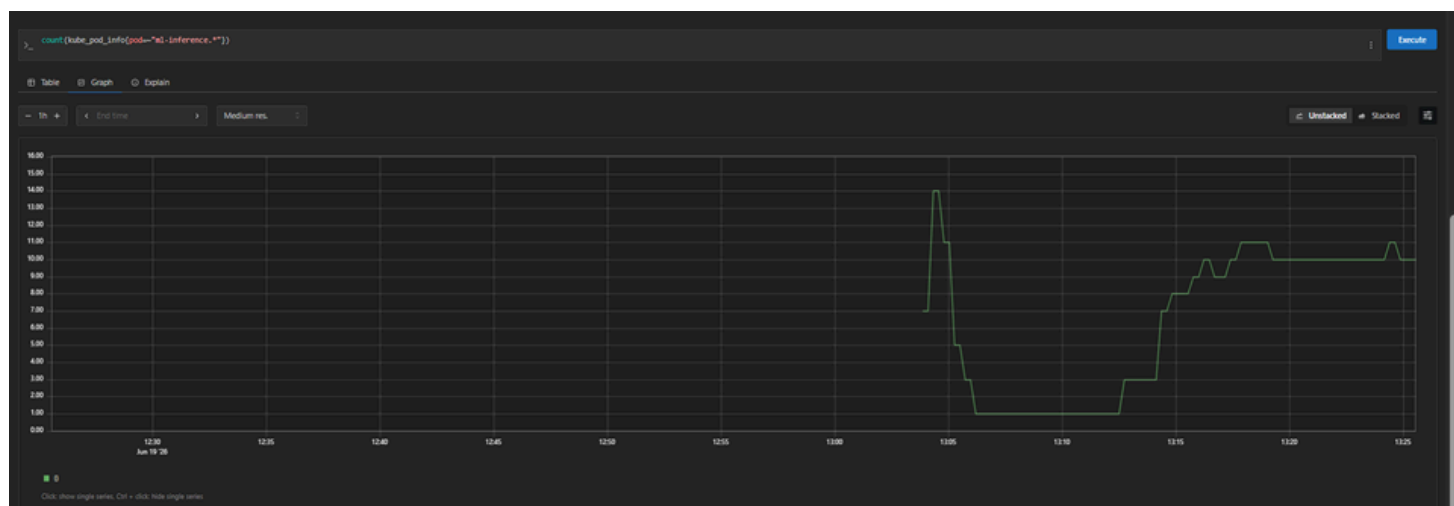


Prometheus Graphs

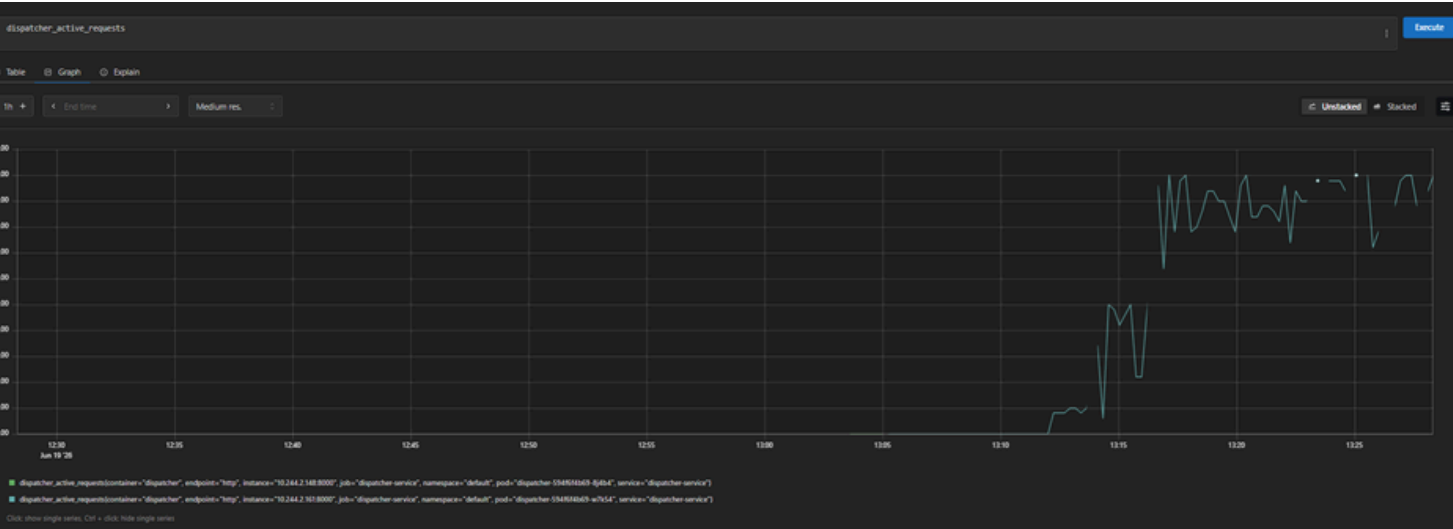
Query - `rate(container_cpu_usage_seconds_total{pod=~"ml-inference.*"}[1m])`



Query - `count(kube_pod_info{pod=~"ml-inference.*"})`



Query - dispatcher_active_requests



Response Time Percentiles

```
(venv) C:\ILMENAU\cc\apl>locust -f locustfile.py --host=http://localhost:8000 --web-port 8091
[2026-06-19 15:11:52,020] LAPTOP-ROLIEI6G/INFO/locust.main: Starting Locust 2.44.1
[2026-06-19 15:11:52,022] LAPTOP-ROLIEI6G/INFO/locust.main: Starting web interface at http://localhost:8091, press enter to open your default browser.
[2026-06-19 15:12:09,469] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 5 users at a rate of 1.00 per second
[2026-06-19 15:12:13,493] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 5} (5 total users)
[2026-06-19 15:13:37,988] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 25 users at a rate of 5.00 per second
[2026-06-19 15:13:41,006] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 25} (25 total users)
[2026-06-19 15:16:04,694] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 50 users at a rate of 10.00 per second
[2026-06-19 15:16:06,704] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 50} (50 total users)
Traceback (most recent call last):
  File "C:\ILMENAU\cc\apl\venv\Lib\site-packages\gevent\ffi\loop.py", line 279, in python_check_callback
    def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument
KeyboardInterrupt
2026-06-19T13:35:02Z
[2026-06-19 15:35:02,609] LAPTOP-ROLIEI6G/INFO/locust.main: Shutting down (exit code 0)
Type Name # reqs # fails | Avg Min Max Med | req/s failures/s
-----|-----|-----|-----|-----|-----|-----|-----|-----|-----
POST /predict 4393 0(0.00%) | 13513 24 35089 13000 | 3.20 0.00
-----|-----|-----|-----|-----|-----|-----|-----|-----|-----
Aggregated 4393 0(0.00%) | 13513 24 35089 13000 | 3.20 0.00

Response time percentiles (approximated)
Type Name 50% 66% 75% 80% 90% 95% 98% 99% 99.9% 99.99% 100% # reqs
-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----
POST /predict 13000 16000 18000 19000 22000 25000 30000 31000 34000 35000 35000 4393
-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----
Aggregated 13000 16000 18000 19000 22000 25000 30000 31000 34000 35000 35000 4393
```

Conclusion :

With a 70% CPU target, the HPA scaled proactively as load increased. CPU utilization exceeded the threshold at 25 concurrent users, triggering additional pods and maintaining 0% request failures up to 50 users. Although latency increased (median from ~3.6s to ~10s), the system remained stable as the autoscaler continuously added replicas to keep CPU usage near the target. This lower threshold improves availability and prevents overload, but requires more resources due to earlier scaling.

```
C:\Windows\System32\cmd.exe x + v -
default namespace

(venv) C:\ILMENAU\cc\apl>kubectl delete pod -l app=ml-inference
pod "ml-inference-6fdb78c657-2sx2s" deleted from default namespace
pod "ml-inference-6fdb78c657-44v7v" deleted from default namespace
pod "ml-inference-6fdb78c657-k5wvr" deleted from default namespace
pod "ml-inference-6fdb78c657-n699b" deleted from default namespace
pod "ml-inference-6fdb78c657-plz5p" deleted from default namespace
pod "ml-inference-6fdb78c657-sb7dz" deleted from default namespace
pod "ml-inference-6fdb78c657-sq4zp" deleted from default namespace

(venv) C:\ILMENAU\cc\apl>kubectl apply -f hpa/hpa-90.yaml
horizontalpodautoscaler.autoscaling/ml-inference created

(venv) C:\ILMENAU\cc\apl>

C:\Windows\System32\cmd.exe x + v -
prometheus-grafana-7455c6bd85-rs64d 3/3
Running 31 (38m ago) 5d2h
prometheus-kube-prometheus-operator-6c65fd7554-hbcbs 1/1
Running 23 (37m ago) 4d2h
prometheus-kube-state-metrics-6cc7c56db5-lwh75 1/1
Running 24 (37m ago) 5d2h
prometheus-prometheus-kube-prometheus-prometheus-0 2/2
Running 25 (5m46s ago) 5d2h
prometheus-prometheus-node-exporter-72xlf 1/1
Running 14 (5m46s ago) 5d2h

(venv) C:\ILMENAU\cc\apl>kubectl get hpa -w
NAME REFERENCE TARGETS MI
NPODS MAXPODS REPLICAS AGE
ml-inference Deployment/ml-inference cpu: <unknown>/90% 1
10 0 12s
ml-inference Deployment/ml-inference cpu: 0%/90% 1
10 1 15s
ml-inference Deployment/ml-inference cpu: 1%/90% 1
10 1 30s
ml-inference Deployment/ml-inference cpu: 0%/90% 1
10 1 45s
|
```

The screenshot shows the Locust web interface in a browser at localhost:8091. The interface displays a table of statistics for the 'locust' host, showing 83 requests, 0 failures, and a current RPS of 0. Below the table, there are tabs for 'STATISTICS', 'CHARTS', 'FAILURES', 'EXCEPTIONS', 'CURRENT RATIO', 'DOWNLOAD DATA', and 'LOGS'. The 'STATISTICS' tab is active.

Below the browser window, there are two terminal windows. The top terminal shows the command to start Locust:

```
(venv) C:\ILMENAU\cc\apl>locust -f locustfile.py --host=http://localhost:8000 --web-port 8091
```

The bottom terminal shows the command to start the Locust runners:

```
(venv) C:\ILMENAU\cc\apl>kubectl port-forward svc/dispatcher-service 8000:8000
```

The output of the bottom terminal shows the port-forwarding process and the handling of connections for the 8000 port.

The screenshot shows the Locust web interface in a browser at localhost:8091. The interface displays the status 'RUNNING' with 25 users, 1.36 RPS, and 0% failures. Below this, there are tabs for STATISTICS, CHARTS, FAILURES, EXCEPTIONS, CURRENT RATIO, and DOWNLOAD DATA. The STATISTICS tab is active, showing a table of performance metrics for two types: /predict and Aggregated. Both show 273 requests, 0 fails, and a median of 7500ms. The /predict type has a 95th percentile of 18000ms and an average size of 36.32 bytes. The Aggregated type has a 95th percentile of 18000ms and an average size of 36.32 bytes.

Below the browser window, a terminal window shows the command prompt output for the Locust command. It indicates that the web interface is starting at http://localhost:8091, and then shows the process ramping up to 25 users at a rate of 5.00 per second. The terminal output is as follows:

```
(venv) C:\ILMENAU\cc\apl>locust -f locustfile.py --host=http://localhost:8000 --web-port 8091
[2026-06-19 16:31:21,361] LAPTOP-ROLIEI6G/INFO/locust.main: Starting Locust 2.44.1
[2026-06-19 16:31:21,364] LAPTOP-ROLIEI6G/INFO/locust.main: Starting web interface at http://localhost:8091, press enter to open your default browser.
[2026-06-19 16:31:57,288] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 5 users at a rate of 1.00 per second
[2026-06-19 16:32:01,324] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 5} (5 total users)
[2026-06-19 16:34:10,766] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 25 users at a rate of 5.00 per second
[2026-06-19 16:34:13,776] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 25} (25 total users)
```

When we increase Load (Users-50, Ram up-10)

C:\Windows\System32\cmd.e

ting Locust 2.44.1
[2026-06-19 16:31:21,364] LAPTOP-ROLIEI6G/INFO/locust.main: Starting web interface at http://localhost:8091, press enter to open your default browser.
[2026-06-19 16:31:57,288] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 5 users at a rate of 1.00 per second
[2026-06-19 16:32:01,324] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 5} (5 total users)
[2026-06-19 16:34:10,766] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 25 users at a rate of 5.00 per second
[2026-06-19 16:34:13,776] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 25} (25 total users)
[2026-06-19 16:35:52,062] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 50 users at a rate of 10.00 per second
[2026-06-19 16:35:54,079] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 50} (50 total users)

locust

localhost:8091

LOCUST

Host: http://localhost:8000, Status: RUNNING, Users: 50, RPS: 2.87, Failures: 0%

STATISTICS

Type	Name	# Requests	# Fails	Median (ms)	90thile (ms)	99thile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
POST	/predict	785	0	11000	43000	47000	12630.97	24	49574	26.86	5.2	0
Aggregated		785	0	11000	43000	47000	12630.97	24	49574	26.86	5.2	0

C:\Windows\System32\cmd.e

Handling connection for 8000
Handling connection for 8000
Handling connection for 8000
Handling connection for 8000
Handling connection for 8000
Handling connection for 8000
Handling connection for 8000
Handling connection for 8000
Handling connection for 8000
Handling connection for 8000
Handling connection for 8000
Handling connection for 8000
Handling connection for 8000
Handling connection for 8000
Handling connection for 8000

C:\Windows\System32\cmd.e

ml-inference Deployment/ml-inference cpu: <unknown>/90% 1
10 6 8m59s
ml-inference Deployment/ml-inference cpu: 16%/90% 1
10 7 9m15s
ml-inference Deployment/ml-inference cpu: 28%/90% 1
10 7 9m32s
ml-inference Deployment/ml-inference cpu: 28%/90% 1
10 7 9m47s
ml-inference Deployment/ml-inference cpu: 30%/90% 1
10 7 10m
ml-inference Deployment/ml-inference cpu: 47%/90% 1
10 8 10m
ml-inference Deployment/ml-inference cpu: 47%/90% 1
10 7 10m
ml-inference Deployment/ml-inference cpu: 18%/90% 1
10 7 10m

C:\Windows\System32\cmd.e

(venv) C:\ILMENAU\cc\apl>kubectl get pods -w

NAME	STATUS	RESTARTS	AGE	READY
alertmanager-prometheus-kube-prometheus-alertmanager-0	Running	23 (20m ago)	5d3h	2/2
custom-autoscaler-759475f877-rl9bv	Running	11 (19m ago)	25h	1/1
dispatcher-594f6f4b69-w7k54	Running	2 (20m ago)	96m	1/1
ml-inference-6fdb78c657-282ms	Running	4 (35s ago)	6m48s	0/1
ml-inference-6fdb78c657-4ql62	Running	6 (45s ago)	9m28s	0/1
ml-inference-6fdb78c657-5b75p	Running	6 (36s ago)	17m	0/1
ml-inference-6fdb78c657-h6fmg	Running	0	102s	0/1
ml-inference-6fdb78c657-hxbf6	Running	0	112s	0/1
ml-inference-6fdb78c657-jf5p9	Running	0	2m2s	0/1
ml-inference-6fdb78c657-mq9qt	Running	4 (24s ago)	6m37s	0/1
ml-inference-6fdb78c657-p7nq2	Running	1 (12s ago)	2m3s	0/1
ml-inference-6fdb78c657-qxwg5	Running	4 (13s ago)	6m25s	0/1
ml-inference-6fdb78c657-rf7qv	Running	4 (34s ago)	6m57s	0/1
prometheus-grafana-7455c6bd85-rs64d	Running	31 (53m ago)	5d3h	3/3
prometheus-kube-prometheus-operator-6c65fd7554-hbcbs	Running	25 (37s ago)	4d3h	1/1
prometheus-kube-state-metrics-6cc7c56db5-lwh75	Running	24 (53m ago)	5d3h	1/1
prometheus-prometheus-kube-prometheus-prometheus-0	Running	25 (20m ago)	5d3h	2/2

C:\Windows\System32\cmd.e

(venv) C:\ILMENAU\cc\apl>kubectl top pods

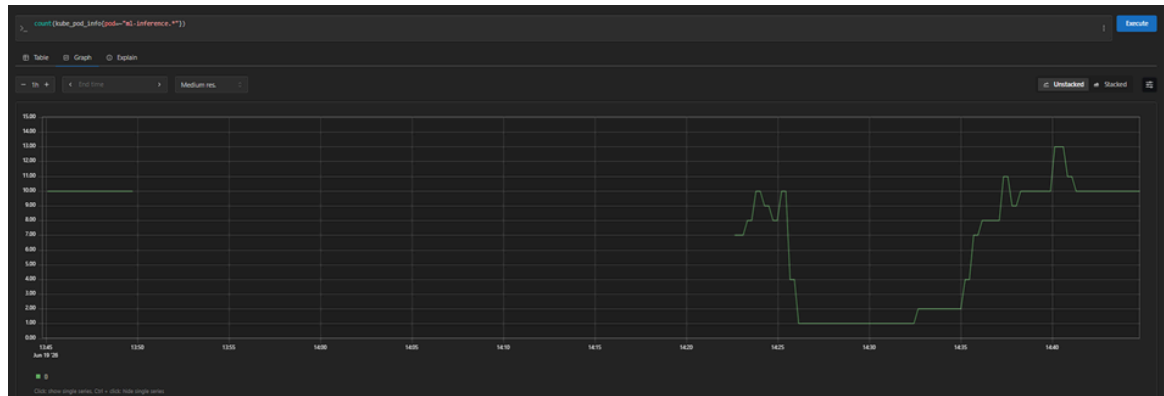
NAME	CPU(cores)
alertmanager-prometheus-kube-prometheus-alertmanager-0	3m
custom-autoscaler-759475f877-rl9bv	4m
dispatcher-594f6f4b69-w7k54	158m
ml-inference-6fdb78c657-282ms	141m
ml-inference-6fdb78c657-5b75p	130m
ml-inference-6fdb78c657-hxbf6	168m
ml-inference-6fdb78c657-jf5p9	94m
ml-inference-6fdb78c657-mq9qt	54m
ml-inference-6fdb78c657-p7nq2	137m
ml-inference-6fdb78c657-qxwg5	111m
ml-inference-6fdb78c657-rf7qv	135m
prometheus-grafana-7455c6bd85-rs64d	16m
prometheus-kube-prometheus-operator-6c65fd7554-hbcbs	10m
prometheus-kube-state-metrics-6cc7c56db5-lwh75	2m
prometheus-prometheus-kube-prometheus-prometheus-0	35m
prometheus-prometheus-node-exporter-72xlf	1m

Prometheus Graphs

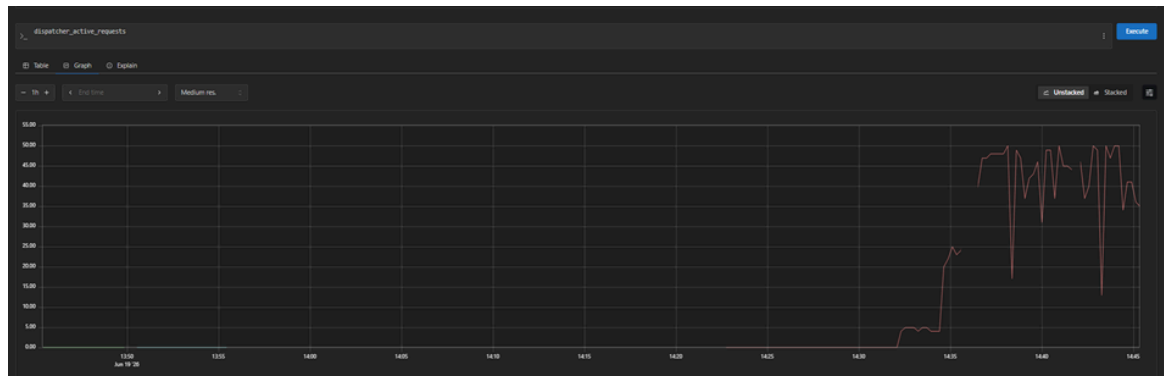
Query - rate(container_cpu_usage_seconds_total{pod=~"ml-inference.*"}[1m])



Query - count(kube_pod_info{pod=~"ml-inference.*"})



Query - dispatcher_active_requests



Response Time Percentiles

```
(venv) C:\ILMENAU\cc\apl>locust -f locustfile.py --host=http://localhost:8000 --web-port 8091
[2026-06-19 16:31:21,361] LAPTOP-ROLIEI6G/INFO/locust.main: Starting Locust 2.44.1
[2026-06-19 16:31:21,364] LAPTOP-ROLIEI6G/INFO/locust.main: Starting web interface at http://localhost:8091, press enter to open your default browser.
[2026-06-19 16:31:57,288] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 5 users at a rate of 1.00 per second
[2026-06-19 16:32:01,324] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 5} (5 total users)
[2026-06-19 16:34:10,766] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 25 users at a rate of 5.00 per second
[2026-06-19 16:34:13,776] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 25} (25 total users)
[2026-06-19 16:35:52,062] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 50 users at a rate of 10.00 per second
[2026-06-19 16:35:54,079] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 50} (50 total users)
Traceback (most recent call last):
  File "C:\ILMENAU\cc\apl\venv\Lib\site-packages\gevent\_ffi\loop.py", line 279, in python_check_callback
    def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument
KeyboardInterrupt
2026-06-19T14:48:02Z
[2026-06-19 16:48:02,739] LAPTOP-ROLIEI6G/INFO/locust.main: Shutting down (exit code 0)
Type      Name      # reqs      # fails | Avg      Min      Max      Med | req/s  failures/s
-----|-----|-----|-----|-----|-----|-----|-----|-----|-----
POST      /predict    3133        0(0.00%) | 12015      24    49573    12000 |    3.27      0.00
-----|-----|-----|-----|-----|-----|-----|-----|-----|-----
Aggregated    3133        0(0.00%) | 12015      24    49573    12000 |    3.27      0.00

Response time percentiles (approximated)
Type      Name      50%      66%      75%      80%      90%      95%      98%      99%      99.9%  99.99%  100% # reqs
-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----
POST      /predict    12000    14000    15000    16000    18000    22000    33000    45000    49000    50000    50000    3133
-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----
Aggregated    12000    14000    15000    16000    18000    22000    33000    45000    49000    50000    50000    3133
```

Conclusion :

The HPA successfully scaled the ml-inference deployment in response to increasing CPU load. At 25 users, performance remained stable, while at 50 users, CPU utilization exceeded the target and triggered scaling from 1 to 2-3 replicas. Prometheus metrics showed higher CPU usage, increased active requests, and corresponding pod scaling. Despite some pod restarts under heavy load, the system continued serving requests without failures, demonstrating effective and reliable autoscaling.

```
(venv) C:\ILMENAU\cc\apl>cd autoscaler

(venv) C:\ILMENAU\cc\apl\autoscaler>kubectl apply -f custom-autoscaler.yaml
error: the path "custom-autoscaler.yaml" does not exist

(venv) C:\ILMENAU\cc\apl\autoscaler>kubectl apply -f deployment_auto.yaml
deployment.apps/custom-autoscaler unchanged

(venv) C:\ILMENAU\cc\apl\autoscaler>kubectl get pods -l app=custom-autoscaler
NAME                                READY    STATUS    RESTARTS    AGE
custom-autoscaler-759475f877-rl9bv  1/1      Running   13 (6m23s ago)  28h
```

The screenshot displays a development environment with two main windows. The top window is a web browser showing the Locust web interface at localhost:8091. The interface includes a status bar at the top indicating the host is 'http://localhost:8000', status is 'RUNNING', and there are 5 users, 1.47 RPS, and 0% failures. Below this, there are tabs for 'STATISTICS', 'CHARTS', 'FAILURES', 'EXCEPTIONS', 'CURRENT RATIO', 'DOWNLOAD DATA', and 'LOGS'. The 'STATISTICS' tab is active, showing a table of request statistics for the endpoint '/predict'. The table has columns for Type, Name, # Requests, # Fails, Median (ms), 95thile (ms), 99thile (ms), Average (ms), Min (ms), Max (ms), Average size (bytes), Current RPS, and Current Failures/s. The data shows two rows: 'POST /predict' and 'Aggregated', both with 924 requests, 0 fails, and a median response time of 1100ms. The bottom window is a terminal running a series of commands. It starts with 'C:\ILMENAU\cc\apl> kubectl port-forward svc/prometheus-kube-prometheus-prometheus 9090:9090', followed by 'Forwarding from 127.0.0.1:9090 -> 9090', 'Forwarding from [::]:9090 -> 9090', and then a series of 'Handling connection for 9090' messages, indicating that the port-forwarding is successful and connections are being handled.

The screenshot displays a development environment with two main windows. The top window is the Locust web interface, showing a 'Host' of 'http://localhost:8000' with a status of 'RUNNING'. It includes a table of statistics for the 'POST /predict' endpoint, showing 994 requests, 0 failures, and a median response time of 1100ms. The bottom window is a terminal running a Kubernetes port-forwarding command: 'kubectl port-forward svc/prometheus-ku-be-prometheus-prometheus 9090:9090'. The terminal output shows the command is successful and lists several 'Handling connection for 9090' messages.

Locust Statistics Table:

Type	Name	# Requests	# Fails	Median (ms)	95thile (ms)	99thile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
POST	/predict	994	0	1100	10000	12000	3118.35	15	14517	46.02	0.9	0
Aggregated		994	0	1100	10000	12000	3118.35	15	14517	46.02	0.9	0

The image displays four windows from a Windows desktop environment:

- Top-Left Window (Web Browser):** Shows the Locust web interface at `localhost:8091`. The interface indicates the Locust instance is **RUNNING** with 50 users, 1.53 RPS, and 0% failures. A table of statistics is visible below the status bar.
- Top-Right Window (Terminal):** Shows a terminal session with the command `(venv) C:\ILMENAU\cc\apl> kubectrl port-forward svc/prometheus-kube-prometheus-prometheus 9090:9090`. The output shows the port-forwarding process starting and handling multiple connections for port 9090.
- Bottom-Left Window (Terminal):** Shows a terminal session with a continuous stream of `Handling connection for 8000` messages, indicating a high volume of connections being handled.
- Bottom-Right Window (Terminal):** Shows a terminal session with the command `kubectl get pods`. The output lists various pods, including `ml-inference-6fdb78c657-ts85m` (Running), `ml-inference-6fdb78c657-dwjsb` (Pending), and `ml-inference-6fdb78c657-ttwzs` (Running).

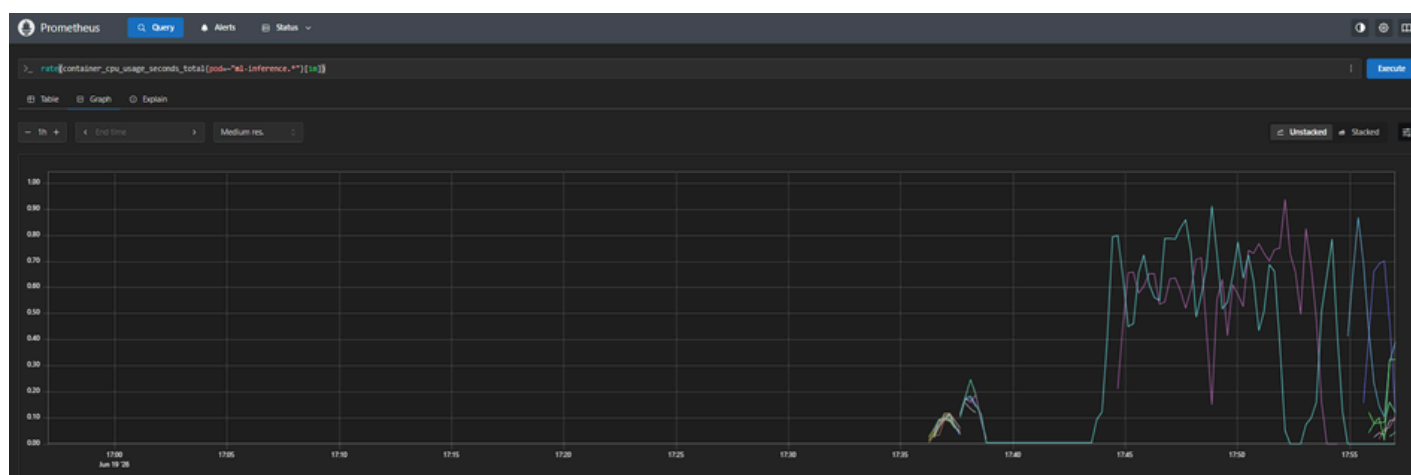
```
C:\Windows\System32\cmd.e x + v
C:\ILMENAU\cc\apl>venv\Scripts\activate

(venv) C:\ILMENAU\cc\apl>kubectl top pods
NAME                                CPU(cores)
MEMORY(bytes)
alertmanager-prometheus-kube-prometheus-alertmanager-0  5m
60Mi
custom-autoscaler-759475f877-rl9bv  2m
84Mi
dispatcher-594f6f4b69-w7k54        114m
89Mi
ml-inference-6fdb78c657-5r4fr       147m
233Mi
ml-inference-6fdb78c657-92x2p       150m
263Mi
ml-inference-6fdb78c657-dwjsb       145m
359Mi
ml-inference-6fdb78c657-gd5kd       141m
308Mi
ml-inference-6fdb78c657-hkhg9       142m
291Mi
ml-inference-6fdb78c657-ts85m       136m
233Mi
ml-inference-6fdb78c657-vq6jp       147m
284Mi
prometheus-grafana-7455c6bd85-rs64d 32m
542Mi
prometheus-kube-prometheus-operator-6c65fd7554-hbcbs  7m
62Mi
prometheus-kube-state-metrics-6cc7c56db5-lwh75  2m
56Mi
prometheus-prometheus-kube-prometheus-prometheus-0  66m
322Mi
prometheus-prometheus-node-exporter-72xlf  7m
24Mi

(venv) C:\ILMENAU\cc\apl>
```

Prometheus Graphs

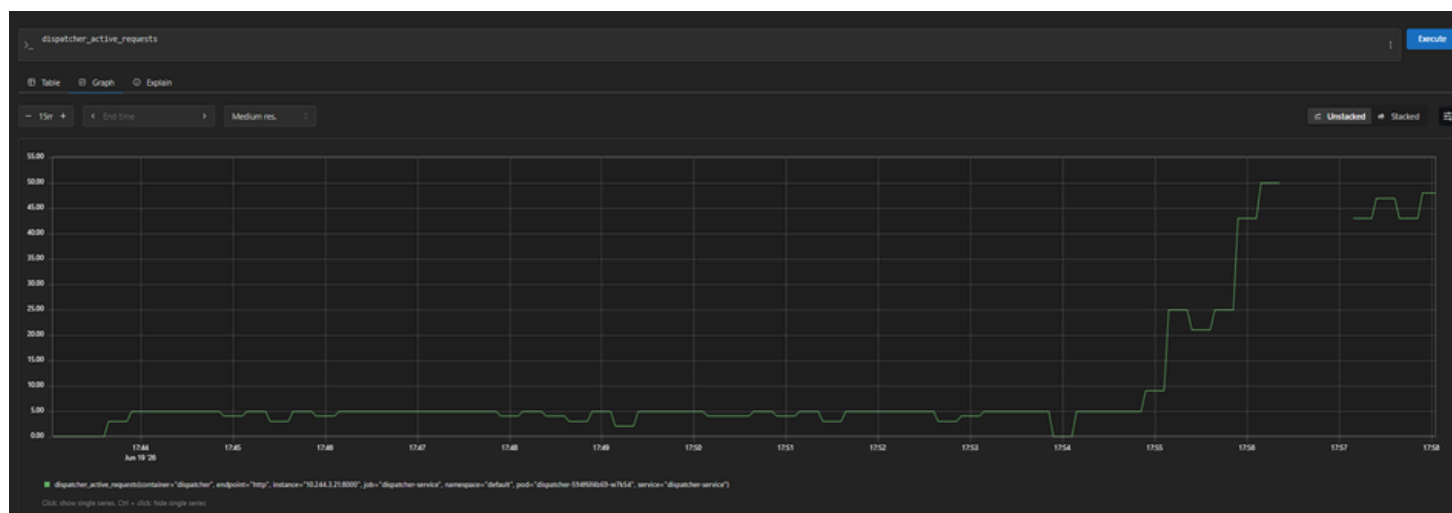
Query - `rate(container_cpu_usage_seconds_total{pod=~"ml-inference.*"}[1m])`



Query - `count(kube_pod_info{pod=~"ml-inference.*"})`



Query - `dispatcher_active_requests`



Response Time Percentiles

```
(venv) C:\ILMENAU\cc\apl>locust -f locustfile.py --host=http://localhost:8000 --web-port 8091
[2026-06-19 19:42:54,881] LAPTOP-ROLIEI6G/INFO/locust.main: Starting Locust 2.44.1
[2026-06-19 19:42:54,884] LAPTOP-ROLIEI6G/INFO/locust.main: Starting web interface at http://localhost:8091, press enter to open your default browser.
[2026-06-19 19:43:33,574] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 5 users at a rate of 1.00 per second
[2026-06-19 19:43:37,690] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 5} (5 total users)
[2026-06-19 19:54:51,165] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 25 users at a rate of 5.00 per second
[2026-06-19 19:54:54,189] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 25} (25 total users)
[2026-06-19 19:55:45,640] LAPTOP-ROLIEI6G/INFO/locust.runners: Ramping to 50 users at a rate of 10.00 per second
[2026-06-19 19:55:47,663] LAPTOP-ROLIEI6G/INFO/locust.runners: All users spawned: {"ImageNetUser": 50} (50 total users)
WARNING: gevent: Unhandled error with no watcher
Traceback (most recent call last):
  File "C:\ILMENAU\cc\apl\venv\Lib\site-packages\gevent\libuv\loop.py", line 39, in python_queue_callback
    def python_queue_callback(self, watcher_ptr, revents):

KeyboardInterrupt
Traceback (most recent call last):
  File "C:\ILMENAU\cc\apl\venv\Lib\site-packages\gevent\ffi\loop.py", line 279, in python_check_callback
    def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument

KeyboardInterrupt
2026-06-19T18:01:24Z
[2026-06-19 20:01:24,561] LAPTOP-ROLIEI6G/INFO/locust.main: Shutting down (exit code 0)
Type Name # reqs # fails | Avg Min Max Med | req/s failures/s
-----|-----|-----|-----|-----|-----|-----|-----|-----|
POST /predict 2175 0(0.00%) | 9467 14 66006 8900 | 2.03 0.00
-----|-----|-----|-----|-----|-----|-----|-----|-----|
Aggregated 2175 0(0.00%) | 9467 14 66006 8900 | 2.03 0.00

Response time percentiles (approximated)
Type Name 50% 66% 75% 80% 90% 95% 98% 99% 99.9% 99.99% 100% # reqs
-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
POST /predict 8900 12000 14000 14000 16000 20000 54000 57000 65000 66000 66000 2175
-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
Aggregated 8900 12000 14000 14000 16000 20000 54000 57000 65000 66000 66000 2175
```

Conclusion :

The custom autoscaler provided the fastest and most workload-aware scaling by reacting directly to active requests and inference latency rather than CPU usage. It scaled earlier and more aggressively than both HPA configurations, increasing pods from 1 to 11-12 under peak load. This proactive behavior maintained 0% request failures at 50 users and kept latency lower than the HPA-90 setup. While it consumed more resources due to the higher pod count, it delivered the best overall responsiveness, stability, and performance for latency-sensitive machine learning inference workloads.

Comparison Table :

Metric	HPA 70%	HPA 90%	Custom Autoscaler
Scaling Trigger	CPU > 70%	CPU > 90%	Dispatcher load / latency / active requests
Reaction Speed	Fast (early scaling)	Slow (late scaling)	Fastest (reacts to real demand)
Scaling Aggressiveness	Moderate	Minimal	High (up to 11-12 pods)
Latency Under Load	Medium	Highest	Lowest / most stable
Failures	None	None	None
Resource Usage	High (more pods)	Low (fewer pods)	Highest (many pods)
Stability	Good	Acceptable but stressed	Excellent
CPU Efficiency	Lower (over-provisioning)	Highest (under-provisioning risk)	Medium
Best For	Balanced workloads	Cost-saving workloads	Latency-sensitive ML inference

Conclusion of the Project :

This project compared three autoscaling strategies for a Kubernetes-based machine learning inference service: HPA with 70% CPU utilization, HPA with 90% CPU utilization, and a custom autoscaler based on active requests and latency.

HPA-70 provided proactive scaling and stable performance but required more resources. HPA-90 was more resource-efficient but delayed scaling, leading to higher latency under heavy load. The custom autoscaler achieved the best overall results by responding directly to workload conditions, maintaining low latency, stable performance, and zero request failures even at peak demand.

Overall, the experiments demonstrate that CPU-based autoscaling alone is not ideal for latency-sensitive ML inference workloads. The custom autoscaler delivered the best balance of responsiveness and reliability, making it the most effective solution despite its higher resource consumption.